Skills

Computational Chemistry:

- Quantum ESPRESSO
- DFT
- Machine Learning
- ORCA
- Molecular Dynamics

Programming:

- Python
- Linux
- Git
- SQL
- Bash
- App Development
- LaTeX
- Pandas
- Matplotlib
- PyTorch
- Numpy
- Scikit-learn

Laboratory Skills:

- FTIR
- X-Ray Spectroscopy
- MOF synthesis
- UV-Vis / DRS
- TGA / DSC

Language Skills

- **Hindi**: Mother tongue
- **English**

Manish Chandra

Date of birth: 26 Jun 2002 | Gender: Male | Nationality: Indian | Mobile phone: +91 7897914523 | Email address: manish.chandra.xy@gmail.com | GitHub: monees007 | Home address: 272175, Khalilabad (India)

Education & Training

Surface Chemistry
13 Jul 2026 - Current

Schrödinger, Inc.

Molecular quantum mechanics, periodic quantum mechanics, and machine learning approaches for studying atomic layer processing and catalysis.

M.Sc. Chemistry
22 Aug 2024 - 30 May 2026

Jawaharlal Nehru University | New Delhi, India

Supramolecular Chemistry, Bioinorganic Chemistry and Drug Discovery, Molecular Spectroscopy, Quantum Chemistry, Organic Chemistry, Inorganic Chemistry, Analytical Techniques in Chemistry, and Group Theory.

Computational Quantum Chemistry
8 Jan 2026 - 30 Apr 2026

NPTEL

Density Functional Theory, Quantum chemistry methods, electronic structure theory, Hartree–Fock, basis sets, molecular properties, ORCA

Diploma in Programming
6 Jan 2021 - 31 Oct 2023

Indian Institute of Technology, Madras | Remote, India

Computer programming and data science foundations with courses in Python, Database Management, Data Structures & Algorithms, Machine Learning Techniques, Tools in Data Science, and project work.

Object-Oriented Programming and Functional Programming in Python
17 Jun 2021 - 31 Jul 2021

Linkedin Learning

B.Sc (Hons.) Chemistry
8 Jul 2019 - 16 Aug 2022

Banaras Hindu University | Varanasi, India

Rigorous honours chemistry curriculum covering core theory (organic, inorganic, physical, analytical), accompanied with biology courses, including genetics, biochemistry, cell biology, etc, with their accompanied laboratory courses.

Publications

Preprint: Phase-Induced Chromatic Transition in [Co(INA)2] MOFs: DFT based study of Gate-Controlled Solvent Interaction

Manish Chandra, Supriya Sabbani | Preprint
Link: | https://figshare.com/s/7b59c8fa3c002b0f1171

Certifications

NET Fellowship - University Grant Commission (Ranked top 6%)

Computer Science and Applications

Qualified Graduate Aptitude Test in Engineering (GATE)

Chemistry

Work experience

Masters Research Scholar

1 Jul 2025 - 20 May 2026

Dr. Supriya Sabbani, Jawaharlal Nehru University

Master's Thesis: Solvent-Modulated Structural Transformations in Cobalt and Copper Coordination Polymers: From Solvatochromic Switching to Hydrogen Evolution Catalysis

Laboratory Teaching Assistant

1 Jun 2024 - 1 Aug 2025

Indian Institute of Technology Madras (Remote) | Chennai, India

Assisted in laboratory sessions, guided students in command-line operations including grep, awk, vim, tmux, ssh, os installation, management and scripting fundamentals for the online laboratory component of course *Introduction to Linux Shell (CS1102)* .

Research Intern

3 Oct 2021 - 18 May 2022

Prof. Satyen Saha, Banaras Hindu University

UV-Visible Spectroscopic Analysis based determination of Isosbestic point of Bromocresol Green.

Recommendations

Masters Research Supervisor

Dr. Supriya Sabbani | ssabbani@jnu.ac.in

Research Mentor

Prof. Satyen Saha | ssaha@bhu.ac.in

Course Instructor

Dr. Dinabandhu Das | jnu.dinu@gmail.com

Extracurricular Activities

Lead, Technical Club

Indian Institute of Technology, Madras

Lead , Technical Club – Coordinated technical events and workshops, fostering peer learning and collaborative problem-solving.

Member, Literary Club

Jawaharlal Nehru University

Contributed to organizing literary and academic events.

Research Projects

Phase-Induced Chromatic Transition in [Co(INA)2] MOFs: DFT based study of Gate Controlled Solvent Interaction

Experimental techniques (DRS, FTIR, TGA) and periodic DFT+U calculations were used to examine the reversible purple-pink to orange color shift in the [Co(INA)] framework. The crystals was changing color upon exposure to moisture or heat (70-120°C), which correlates with the movement of guest molecules (like DMF and water) in and out of the material's pores. The computed HOMO-LUMO gaps across five polymorphs appear to align with the experimental color shifts (497 nm to 533 nm) observed via DRS. PDOS analysis suggests that this color change is likely driven by the rotation of the ligand's pyridyl rings when interacting with guest molecules, rather than a transition at the cobalt metal center itself.

Study of Solvent-Induced Redox Reactions in Copper Coordination Polymer: Catalysis,and Electrochemical Hydrogen Evolution.

We studied solvent-induced redox transformations in copper-based coordination polymers, including $[Cu(INA)Cl_3]_n$ and $[Cu(4,4'\text{-bipy})Cl_3]_n$. It was observed that exposure to coordinating solvents triggers a solid-to-solid phase transformation, seemingly driven by the oxidation of the metal centers from Cu(I) to Cu(II). This structural change could be temporarily reversed using an aqueous solution of ascorbic acid and chloride ions. Furthermore, the study evaluates these copper complexes for electrocatalytic hydrogen evolution. Preliminary linear sweep voltammetry indicates that the $[Cu(H2O)_2(4,4'\text{-bipy})_4]$ complex might achieve a benchmark current density of 10 mA/cm² at an overpotential of 0.134 V.

Choosing Dopant in Copper Coordination Polymer for Hydrogen Evolution Reaction using free energy calculation by DFT.

This project investigates the effect of dopant selection on the catalytic performance of a copper-based coordination polymer for the hydrogen evolution reaction (HER), using density functional theory (DFT) to compute the Gibbs free energy of hydrogen adsorption (G_{H^*}) at various active sites.